

SE200 Oil Well Drive
Version 1.0
Operations Manual

SOFTWARE VERSION 1.6

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Keypad (HMI)

The main screen of the SE200's HMI includes several buttons and indicators to provide feedback and control of the VFD. An example of the main screen is shown in the figure below.

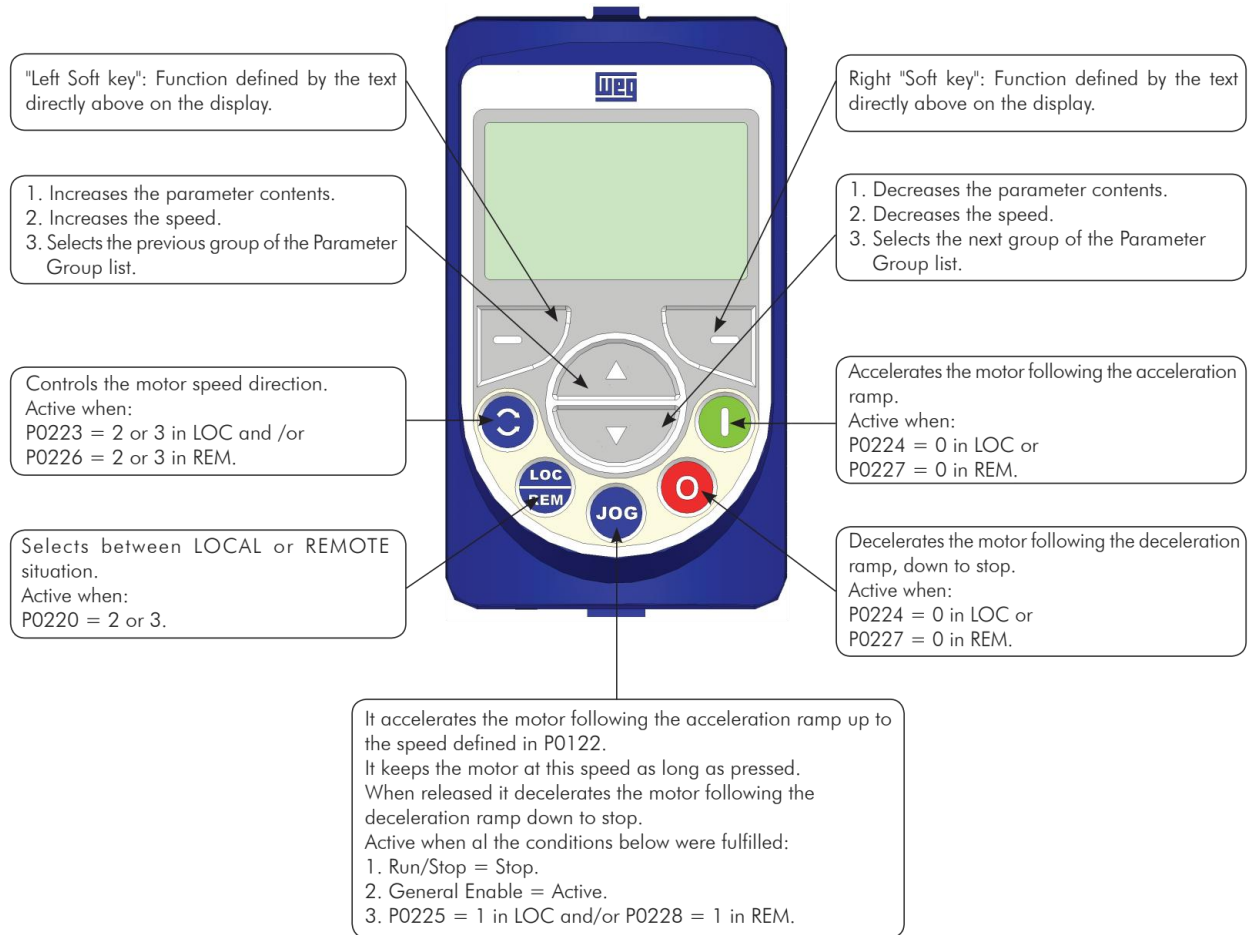


Figure 1 Keypad

Power and Motor Terminations

The SE200 VFD supports both single-phase and three-phase applications. When using it in a single-phase setup, follow these key guidelines to ensure proper operation:

1. Power Input Connections

- Connect **single-phase** power to A/L1 (**Red**) and B/L2 terminals (**Black**). Connect **three-phase** power to A/L1 (**Red**), B/L2 terminals (**Black**) and C/L3 terminals (**Blue**).
 - These terminals supply power to the DC bus of the CFW11.

2. Motor Wiring

- Terminate the motor leads at the U, V, and W terminals, located at the bottom right of the VFD.

3. Sizing Requirement

- When using the SE200 in single-phase mode, select a VFD rated at twice the intended motor load.
- This ensures sufficient capacity and reliable performance.

4. Parameter Configuration

- Set P0357 = 0 sec to bypass the phase loss fault (this is required for operation).

Following these steps will help ensure efficient and reliable operation of the SE200 VFD in a single-phase configuration

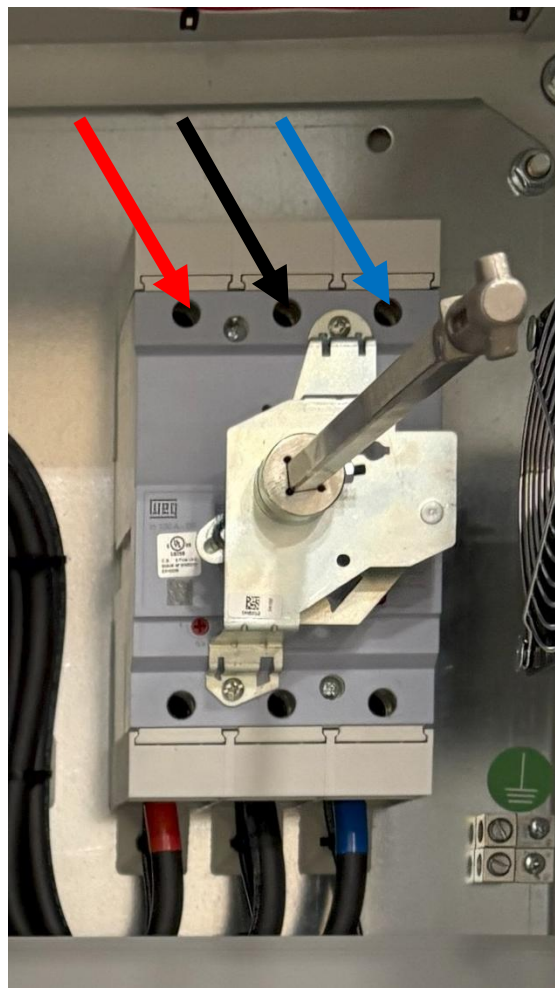


Figure 2 Main Breaker

Control Wiring

For external shutdowns, such as Prescos, Vibration Switches, the control wiring can be connected as outlined below.

1. Jumper Removal

- Remove jumpers only for the specific shutdowns that are required.

2. Alarm Indications

- The following alarms will appear on the HMI for corresponding shutdowns:
 - A750: Shutdown 1
 - A751: Shutdown 2
 - A752: Shutdown 3

3. Additional Information

- Refer to the wiring diagram for detailed instructions and additional guidance.

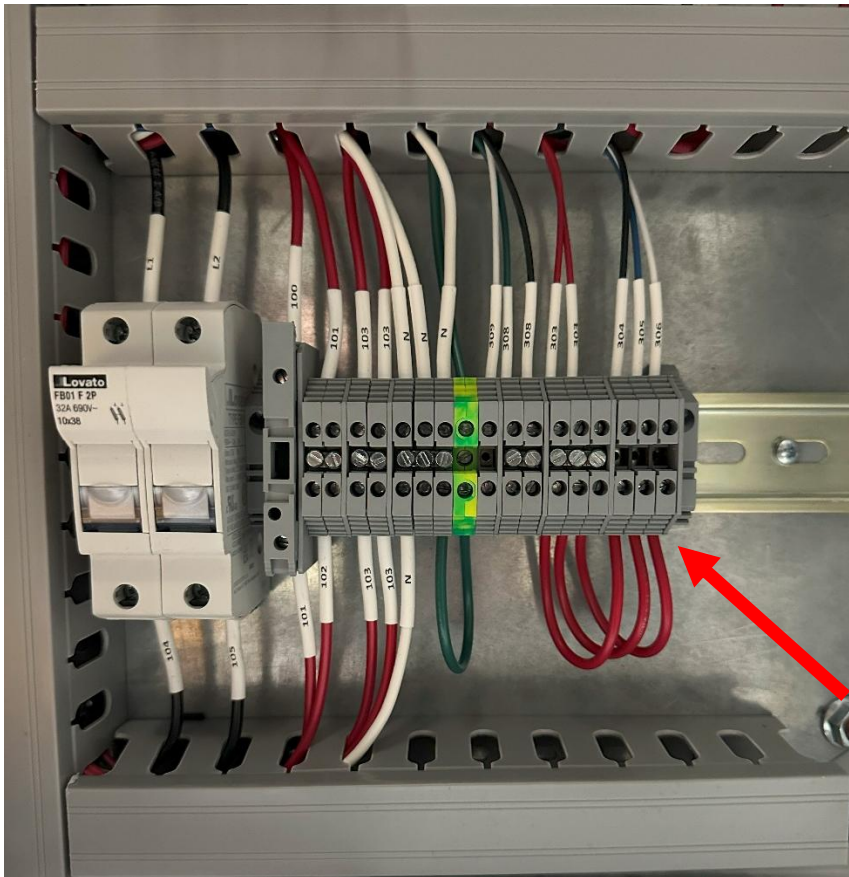


Figure 3 Basic terminal strip layout

Initial Power Up

When powering up the SE200 for the first time, the field technician must configure the device using the **Field Technician Parameters** section outlined in our startup guide. These parameters are detailed below, and a full copy of the guide is included with every **SE200** unit. **Parameter steps #1-#32 are set in SEEL factory and do not need to be set in the field unless otherwise told by SEEL Representative, or in other parts of this manual.**

Steps to Access and Configure Parameters:

1. Access the Parameter Section

- Press **Menu > Select All Parameters**.
- Use the **Up** and **Down** arrows to navigate to parameter **P0202**.

2. Complete the Oriented Startup Menu

- Starting at Step 33, the SE200 VFD will guide you through the oriented startup steps (**Steps 33–41**).
- After completing the auto-tune process, the device will return to the main VFD screen.

3. Continue Programming

- Return to **Menu > All Parameters** to proceed with the programming procedure (**Steps 42–52**).

<u>PARAMETERS TO BE SET BY SEEL FACTORY (Steps #1-32)</u>					
Step#	Parameter #	Parameter Description	Unit	Default Value	Set Value
1	P0000	Access to Parameters	0-9999	0000	0005
2	P0100	Acceleration Time	Sec	20.0	5.0
3	P0101	Deceleration Time	Sec	20.0	5.0
4	P0132	Max Over Speed Level	0-100%	10%	50%

5	P0205	Reading Parameter Selection #1 (Display in SPM)	Speed Reference Unit	2	1
6	P0209	Reference Engineering Unit #1	Strokes Per Minute	R	S
7	P0210	Reference Engineering Unit #2	Strokes Per Minute	P	P
8	P0211	Reference Engineering Unit #3	Strokes Per Minute	M	M
9	P0212	Reference Decimal Point	wxyz	0	1 (wxy.z)
10	P0217	Zero Speed Disable	Disable Magnetization Current	0	1
11	P0220	LOC/REM Selection Source	LR Key LOC	2	2
12	P0221	LOC Reference Select	AI1	0	1
13	P0222	Remote Reference Select	Serial / USB	0	9
14	P0223	LOC FWD/REV Selection	Always FWD	2	0
15	P0224	LOC Run/Stop Selection	Soft PLC	0	5
16	P0225	LOC JOG Selection	Disable	1	0
17	P0226	REM FWD/REV Selection	Always FWD	4	0
18	P0227	REM Run/Stop Selection	Serial / USB	1	2
19	P0228	REM JOG Selection	Serial / USB if using RS485-01 in Slot 3, Any bus-CC If using Modbus/TCP or Ethernet card in slot 4	2	3 or 4

20	P0229	Stop Mode Selection	Coast to Stop	0	1
21	P0233	AI1 Signal Type	0-10vdc (0)	0	0
22	P0263	DI1 Function	Not Used (Spare)	1	0
23	P0264	DI2 Function	PLC Use (Shutdown #1)	8	21
24	P0265	DI3 Function	PLC Use (Shutdown #2)	0	21
25	P0266	DI4 Function	PLC Use (Shutdown #3)	0	21
26	P0267	DI5 Function	Not Used (Spare)	10	0
27	P0268	DI6 Function	General Enable	14	2
28	P0275	DO1 Function	Soft PLC	13	28
29	P0276	DO2 Function	Run	2	11
30	P0277	DO3 Function	No Fault/Alarm	1	36
31	P0320	Fly Start/Ride-Through	Ride-Through	0	3
32	P0353	IGBT's / Air Overtmp Config	Under Temp Disable	0	4
<u>PARAMETERS TO BE SET BY FIELD PERSONEL (Steps #33-52)</u>					
33	P0202	Type of Control	Sensorless Vector	0	3
34	P0296	Line Rated Voltage	0-6 (Can be adjusted if overvoltage or undervoltage trips occur)	4- (480)	Set as per input voltage
35	P0398	Motor Service Factor	As per motor nameplate		Motor SF
36	P0400	Motor Rated Voltage	As per motor nameplate		Motor Voltage
37	P0401	Motor Rated Current	As per motor nameplate		Motor Current
38	P0402	Motor Rated Speed	As per motor nameplate		Motor RPM

39	P0403	Motor Rated Frequency	As per motor nameplate		Motor Hz
40	P0404	Motor Rated Power	As per motor nameplate		Motor HP/KW
41	P0408	Run Self Tuning	No Rotation (if "Close General Enable" is displayed ensure on/off switch is on)	0	1
42	P0156	Overload Current 100% Speed	0.1 to 1.5 x Inom-ND Example: Motor FLA – 13.2A If SF is 1.0, P0156 should be set to 15.2A If SF is 1.15, P0156 should be set to 16.5A	1.05 x Inom-ND	As per motor service factor, adjust if required (SF=1.0 – 1.15 x Inom-ND) (SF=1.15 – 1.25 x Inom-ND)
33	P0157	Overload Current 50% Speed	0.1 to 1.5 x Inom-ND DO NOT ADJUST UNLESS REQUIRED	0.9 x Inom-ND	Adjust if required
43	P0158	Overload Current 5% Speed	0.1 to 1.5 x Inom-ND DO NOT ADJUST UNLESS REQUIRED	0.65 x Inom-ND	As per motor service factor, adjust if required
44	P0159	Motor Thermal Class	Class 5,10,15,20,25,30,35,40,45	1 – Class 10	3 – Class 20
45	P0169	Max + Torque Current	0-350% (Adjust if required)	125%	150%
46	P0170	Max – Torque Current	0-350% (Adjust if required)	125%	150%

47	P0185	DC Link Regulation Level	Adjustable Voltage Range as per input voltage	800V	770V
48	P0186	DC Link Proportional Gain	0-63.9	18	40
49	P0325	Ride-Through P Gain	0.0-63.9	22.8	10.0
50	P0340	Auto Reset Time (VFD Faults)	0-3600s	0	Set to desired reset time
51	P0357	Line Phase Loss Time	0-60sec	3 sec	Set to 0 seconds
52	P1001	Soft PLC Command	Ensure this is set to "Run Program"	0	1

Table 1 Startup guide

Pumpjack Information

Steps 57-77: Pumpjack Information and Configuration

To program steps 22-42, you must gather specific pumpjack details. The technician will need to obtain the following:

1. **Motor sheave size (inches)**
2. **Gearbox sheave size (inches)** – This may be stamped on the side of the shiv.
3. **Pumpjack gearbox ratio** – Typically stamped on the gearbox nameplate (commonly 30:1).

Programming Steps Breakdown:

- **Steps 57-63:** Highlighted in **yellow**, these steps are required for the operation of the SE200.
- **Steps 64-77:** Highlighted in **green**, these steps are optional and not necessary for basic operation.

Setting Parameters:

1. After gathering the required pumpjack information, enter the parameters and reference the calculated value in **P1013**.
2. Use the value in P1013 as the base to set **P1017**.

Adjusting Parameter P1017:

- Set P1017 to match the value in P1013 initially.
- Start the pumpjack and use the speed dial to set the **HMI to 3.0 SPM**.
- Using a timer, check that one revolution takes **exactly 20 seconds** at 3.0 SPM:
 - If the time is too high or low, adjust **P1017** up or down as needed.
 - After adjusting, reset the speed pot to 3.0 SPM on the HMI and time the stroke again.
 - Repeat the process until one revolution is consistently 20 seconds.

Additional Notes:

- On new wells, after the pumpjack has pumped off fluid and weights have been adjusted, **P1017** may require further adjustments.

57	P1010	Minimum Strokes Per Minute	0-15	0	Desired Min SPM
58	P1011	Maximum Strokes Per Minute	0-15	0	Desired Max SPM
59	P1012	Motor RPM	0-1800	0	Motor RPM
60	P1014	Motor Sheave Size (inches)	0-18	0	Motor Sheave (in)
61	P1015	Pumpjack Gear Box Ratio	0-100 (typical is 30:1)	0	30
62	P1016	Pumpjack Sheave Size (in)	0-100	0	Unit Sheave (in)
63	P1017	Reference Scale Factor	After all the data is entered, Set P1017 to value displayed in P1013 (value can be adjusted up or down slightly to calibrate SPM on VFD HMI to actual running SPM)	0	0-200 as per value displayed in P1013
64	P1020	Shutdown #1 Debounce Timer (Sec)	Set to desired shutdown debounce time up to 5sec	0	Recommend leave at 0
65	P1021	Shutdown #2 Debounce Timer (Sec)	Set to desired shutdown debounce time up to 5sec	0	Recommend leave at 0

66	P1022	Shutdown #3 Debounce Timer (Sec)	Set to desired shutdown debounce time up to 5sec	0	Recommend leave at 0
67	P1024	Well On/Off Timer Enable	Set to 1 if on/off Timer is to be used	0	0 – Disable 1 – Enable
68	P1025	Well On Time (Mins)	Set to desired On run time	0	Desired run mins
69	P1026	Well Off Time (Mins)	Set to desired Off run time	0	Desired run mins
70	P1027	Start Up Delay Enable	Set to 1 to Enable Start Up Delay	0	Set to 1
71	P1028	Start Up Delay (Sec)	Set to Start Delay Time in Seconds (0-32767 sec)	0	Set to desired time
72	P1029	Shutdown Auto Restart Enable	Set to 1 to enable Shutdowns to Auto Reset as per settings in P1030 & P1031	0	Set to 1
73	P1030	Shutdown Auto Restart Attempts Before Unit Stays Locked Out	Set to desired amount of Auto Restart attempts (0-10 Attempts), Shutdown trip counter will reset after 10 minutes of consecutive run time	0	Set to desired restart attempts
74	P1031	Shutdown Auto Restart Wait Time (Sec)	Set to desired wait time between Auto Restart Attempts in Seconds (0-32767 sec)	0	Set to desired wait time
75	P1033	Belt Break Torque Shutdown %	Set to desired Torque Shutdown % (typically set to 5%) Shutdown is enabled after a 30 second start up delay timer	0	Set to desired torque SD %
76	P1034	Belt Break Delay (Sec)	Set the desired time delay for motor torque to be consistently below setpoint in P1033 (0-600 sec, typically set to 15sec)	0	Set to desired time
77	P1035	Powerless Restart	When enabled the unit will automatically restart after power loss (if on/off switch is on)	0	0 – Disable 1 – Enable

Table 2 Pumpjack parameters

Dynamic Brake (Optional)

A **Dynamic Braking Resistor (DBR)** is generally not required for the SE200. However, when necessary, the DBR must be appropriately sized for the specific application.

DBR Requirements Based on Application

1. **For SE200 Units 75 HP and up**
 - A braking unit is required and must be wired in parallel with the DBR (refer to **Figure 4**).
 - Connect wires to the **DC+** and **DC-** to the input + and -. From the braking unit run wires from the output + and – to the braking resistor.
 - It is important to note that the braking unit must be set to 460 on the jumper.
 - For multiple chokes:
 - Set the first choke to **Master Mode**.
 - Set all additional chokes to **Slave Mode**.
 - Interconnect the chokes using a twisted pair of wires.
2. **For SE200 Units 60 HP and Less**
 - After fitting the DBR to the SE200 cabinet:
 - Connect the brake to the **DC+** and **BR** terminals.
 - Configure **Parameters P0153 to P0155**
 - **P0153-Dynamic Braking Level** this is set based off your **P0296** setpoint
 - **P0154-Dynamic Braking Resistor** set this to the size of the DBR that has been installed.
 - **P0155-Dynamic Braking Resistor Power** set this the rated DBR power.

(Note: These parameters only need to be configured for VFDs rated at 60 HP or less.)

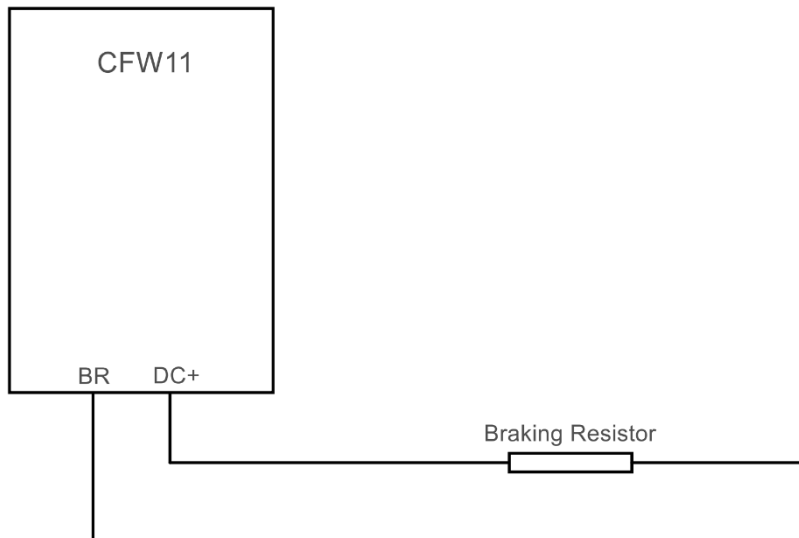


Figure 4 60HP and Less DBR wiring

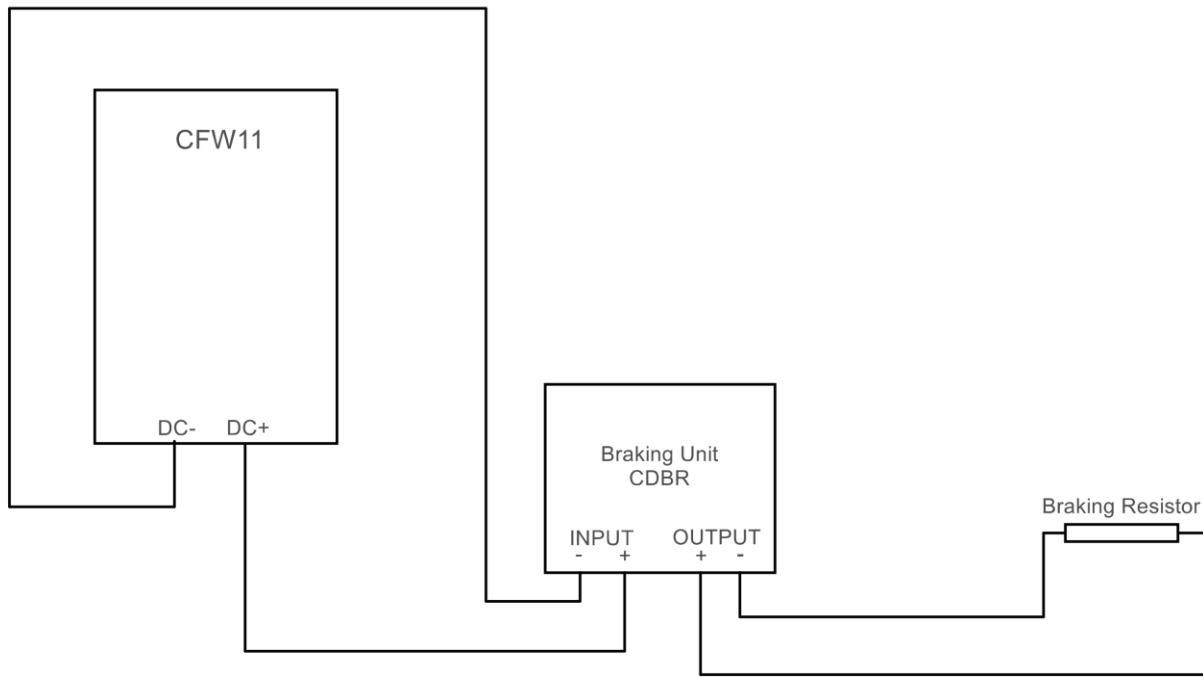


Figure 4a 75HP and Up DBR wiring

Troubleshooting

For common faults/alarms and their possible causes see table starting on page 13.

Faults/Alarms	Description	Possible Causes
F006 Imbalance or Input Phase Loss	Mains voltage imbalance too high or phase missing in the input power supply. Note: - If the motor is unloaded or operating with reduced load this fault may not occur. - Fault delay is set at parameter P0357. P0357 = 0 disables the fault. - If single-phase power supply is used, this fault must be disabled.	Phase missing at the inverter's input power supply. Input voltage imbalance >5 %. For the Frame Size E: Phase loss at L3/R or L3/S may cause F021 or F185. Phase loss at L3/T will cause F006. For frame sizes F and G: Pre-charge circuit fault.
F021 DC Link Undervoltage	DC Link undervoltage condition occurred	The input voltage is too low, and the DC bus voltage dropped below the minimum permitted value (monitor the value at parameter P0004): DC BUS Voltage < 446 V - For a 440-460 V input voltage (P0296 = 3). DC BUS Voltage < 487 V - For a 480 V input voltage (P0296 = 4). Phase loss in the input power supply. Pre-charge circuit failure. Parameter P0296 was set to a value above of the power supply rated voltage. When free spooling can increase P0169 and P0170 as needed depending on speed.

F022 DC Link Overvoltage	DC link overvoltage condition occurred.	The input voltage is too high, and the DC bus voltage surpassed the maximum permitted value: DC Bus Voltage > 400 V - For 220-230 V input models (P0296 = 0). DC Bus Voltage > 800 V - For 380-480 V input Inertia of the driven load is too high, or deceleration time is too short. Wrong settings for parameters P0151, or P0153, or P0185. Ensure P0185 and P0186 are set to 770 and 40 respectively. When free spooling can increase P0169 and P0170 as needed depending on speed.
F048 IGBT Overload Fault	An IGBTs overload fault occurred.	Inverter output current is too high.
F070 Overcurrent / Short-circuit	Overcurrent or short-circuit detected at the output, in the DC bus, or at the braking resistor.	Short-circuit between two motor phases. Short-circuit between the connection cables of the dynamic braking resistor. IGBT modules are shorted. Disconnect Motor try running VFD if issue persists replace inverter, if problem goes away check motor and cable for shorts perform a megger test.
F071 Output Overcurrent	The inverter output current was too high for too long.	Excessive load inertia or acceleration time too short. Settings of P0135 or P0169 and P0170 are too high
F072 Motor Overload	The motor overload protection operated. Note: It may be disabled by setting P0348 = 0 or 3	Settings of P0156, P0157 and P0158 are too low for the used motor. Motor shaft load is excessive.
F0150 Motor Overspeed	Overspeed fault. It is activated when the real speed exceeds the value of $P0134 \times (100 \% + P0132)$ for more than 20 ms.	Wrong settings of P0161 and/or P0162. Problem with the hoist-type load. Adjust P0132 higher or increase P01011.

A163/164 Break Detect AI1/AI2	It indicates that the AI current (4-20 mA or 20-4 mA) reference is out of the 4 to 20 mA range.	Broken AI cable. Bad contact at the signal connection to the terminal strip. Blown control fuse on terminal strip for AI circuit. If there is no AI going to that input set the corresponding input P0233-AI1 or P0238-AI2 to 0.
A181 Invalid Clock Value	Invalid clock value alarm	It is necessary to set date and time at parameters P0194 to P0199. (Try this first if not then check battery) Keypad battery is discharged, defective, or not installed.
F185 Pre-charge Contactor Fault	It indicates fault at the pre-charge contactor	Pre-charge contactor defect. Open command fuse. Phase loss in the input L1/R or L2/S. P0355 = 1 (Incorrect setting for model's frame E supplied by the DC link. For those models, the setting should be P0355 = 0.
A750 External Shutdown #1	Indicates an external shutdown is tripped	Check Presco, check vibration switch, check pollution switch, ensure jumpers left are tight and secure. Ensure 24VDC loop is complete.
A751 External Shutdown #2	Indicates an external shutdown is tripped	Check Presco, check vibration switch, check pollution switch, ensure jumpers left are tight and secure. Ensure 24VDC loop is complete.
A752 External Shutdown #3	Indicates an external shutdown is tripped	Check Presco, check vibration switch, check pollution switch, ensure jumpers left are tight and secure. Ensure 24VDC loop is complete.

"Close General Enable" won't go away and drive wont autotune during setup	Soft PLC is not set to run program or loose/incorrect wiring	Ensure Soft PLC Command P1001 is set to "Run Program" Verify wiring is correct and connections are tight. Press Green button, ensure on/off switch is in the on position Ensure all S/D cables are connected and not tripped. Switch parameter P0202 back to V/F 60hz and ensure "Ready is displayed in top left-hand corner"
VFD wont run after power bump/loss	When the on/off switch is on, but the VFD does not run	Ensure that the green HMI button has been pressed. Alternatively, if you want the VFD to restart automatically Parameter P1035 can be enabled for power loss restart
VFD stuck in auto tuning menu	Auto tune not completed	Ensure auto tune has been completed scroll down to bottom to P0408 to "Run self tuning?" set to 1 for "No Rotation" turn hand switch on and press green button

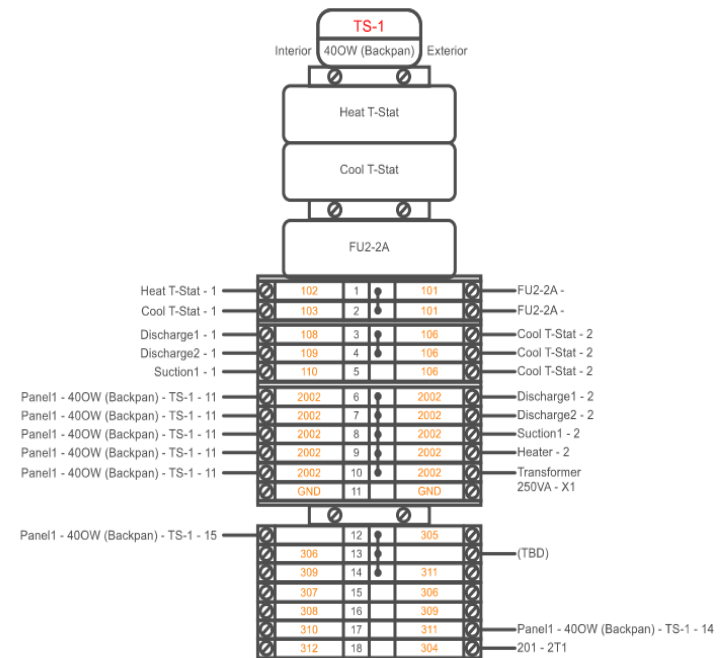
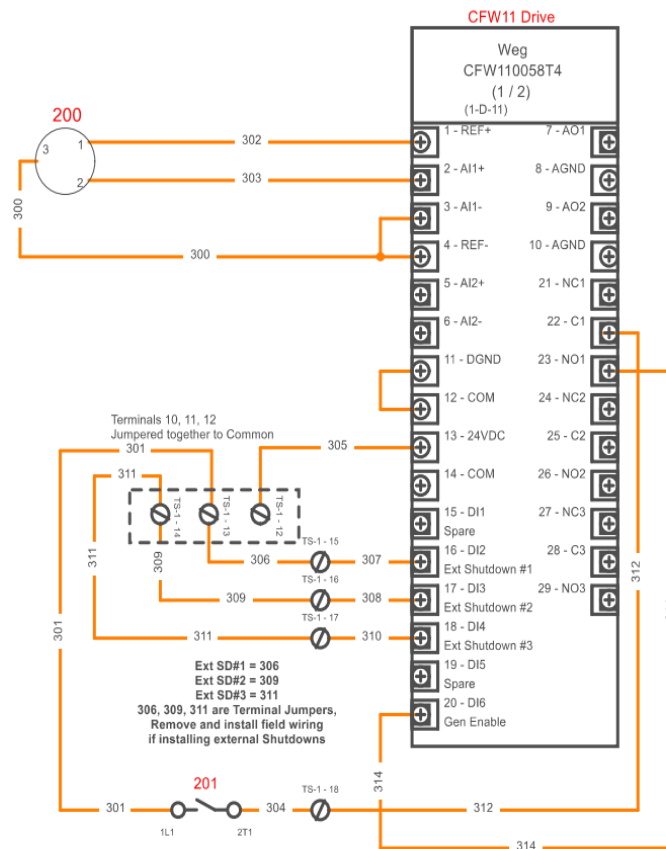


Figure 5 Typical Control Wiring Diagram

For CFW11 Alarm & Fault List refer to page 6-2 of the CFW-11 User Manual Provided

SEEL Software Alarm List:

A750 – External Shutdown #1
A751 – External Shutdown #2
A752 – External Shutdown #3
A754 – Start Up Delay
A755 – Belt Break Shutdown
A799 – Well Off Time

Analog Input / Output Configuration: (Dipswitch's Located at top of control board)

S1.1 (Analog Output 1): Dip Switch Off - 4-20mA, Dip Switch On: 0-10vdc
S1.2 (Analog Output 2): Dip Switch Off - 4-20A, Dip Switch On: 0-10vdc
S1.3 (Analog Input 2): Dip Switch Off – 0-10vdc, Dip Switch On: 4-20mA
S1.4 (Analog Input 1): Dip Switch Off – 0-10vdc, Dip Switch On: 4-20mA

Notes:

- Red Button on Keypad is utilized to reset alarms/faults, this will also reset the on/off timer.
- Green Button on Keypad is utilized to start VFD in local.
- If equipped red beacon illuminates when shutdown #1, # 2, 3 are tripped or when a VFD alarm or fault has occurred.

Digital Inputs:

- DI1 – Not Used (Spare)
- DI2 – Shutdown #1 Connection (Normally Closed), this shutdown can be programmed for auto restart
- DI3 – Shutdown #2 Connection (Normally Closed), this shutdown can be programmed for auto restart
- DI4 – Shutdown #3 Connection (Normally Closed), this shutdown can be programmed for auto restart
- DI5 – Not Used (Spare)
- DI6 - General Enable (Normally Open)

Technical Support

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Notes: